

IFN006 Research Information Strategy Report

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Research question: How to build a MARL simulation, parametrized by preference vectors, to predict consensus output?

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1. Introduction

This document presents a detailed Research Information Strategy Report subdivided into sections. It starts with the research question. Then, this question is decomposed into concepts and keywords that will help construct the search string. It shows the database selection followed by the search strategy that contains the search statement and its findings. Afterwards, it presents the information management strategy followed by the current awareness strategy, where the document will show how to keep the researcher updated with the latest developments in the field. Finally, it will move to the analysis of some sources; in this section, three provided sources are analysed for authenticity.

2. Research question

2.1. Research question

The following research question is part of a set of three that is the actual authors research project. For this document, only the third one is analysed.

Research question: How to build a MARL simulation, parametrized by preference vectors, to predict consensus output?

This project uses natural language processing to analyse diplomatic text that has been submitted to the Antarctic Treaty Consultative Meetings (ATCM) that take place every year. The first and second parts of this project contain two questions that are focused on extracting, out of the diplomatic text, the positions or vectors of different countries throughout the history of the ATCMs. These are the parametric vectors that, in the question above, are used to build a multi-agent reinforcement learning (MARL) simulation, to match the historical outputs of the ATCM, and then predict the outputs of future ATCMs, which decide everything through consensus.

2.2. Research concepts

To build a structured search string, the research question is deconstructed into research concepts. These concepts are taken apart further into keywords. The research concepts that were found in this question are the following:

- MARL simulation.
- Parametric preference vectors.
- Consensus output prediction.

Three research concepts span three different fields that define the scope of the literature search. The first one is computer science, e.g., deep learning. The second one is game theory and negotiation theory. And the third one is consensus dynamics and prediction, which belong to a broader area of decision science.

In the following section, the document shows a set of synonyms, related terms, and keywords that are used for the search string in the database.

2.3. Keywords, synonyms and related terms

To find the synonyms and related terms that researchers use when publishing relevant studies to the literature review, the research concepts are deconstructed into keywords. These keywords are the second column in Table 1. The first column shows the concept that these keywords were derived from, and the third and fourth columns show the related terms or synonyms that are used for our literature search.

Table 1. Keywords

Research concept	Keyword	Synonyms	Related terms
MARL simulation	Multi-agent reinforcement learning (MARL)	multi-agent RL; decentralised reinforcement learning	agent-based simulation; Markov/stochastic games; multi-agent systems
Parametric preference vectors	Preference vector	preference weights; preference-conditioned policy	multi-objective reinforcement learning (MORL); scalarisation; reward weighting
Consensus output prediction	Consensus	Absolute agreement; aggregated output	consensus algorithms; ensemble prediction; voting/aggregation; Unanimity

3. Database

Considering that the research concepts span three different fields: computer science for agent simulations and parametrized vectors, game theory and decision science literature for consensus dynamics, the selected database needs to have access to all these fields to avoid missing relevant information.

Considering the above criteria, the selected database is **Scopus**. This database was selected using five different criteria:

- The first one is comprehensiveness. It contains more than one hundred million records across three hundred and thirty subject areas. It also contains all the needed literature from different areas: simulation, algebra, game theory, and decision science.
- The second criterion is scope and coverage. It is multidisciplinary and cross-publisher. It also indexes IEEE, ACM, Springer, Elsevier, and other social sciences publishers.
- The third criterion is quality. All indexed papers are peer-reviewed and curated.

- The fourth criterion is the search interface, syntax, and the search algorithm. It facilitates all the Boolean and field-targeted searching and filtering needed.
- The fifth criterion is usability. It is easy to access since it is web-based.

4. Search strategy

4.1. Formulate search statement

The initial goal is to build the search statement using Table 1, going row-wise (i.e. per research concept). Per row, the keywords, synonyms and related terms of the first research concept are joined using OR Boolean operators, except when the terms are necessarily mutually exclusive. The same applies to the other table rows. The resulting strings from each table row are joined together using the AND Boolean operator to restrict the results to interdisciplinary studies, and are applied as field search on title, abstract and keywords.

Since there are several multi-word terms, phrase searching was applied in several places of the search statement, for example, "MARL simulation" and "multi-objective reinforcement". Furthermore, nesting is used to group synonyms and truncations for the words that commonly vary in form, for example, "reinforc*", which can take the form of "reinforcement" or "reinforced".

The resulting search statement is:

```
TITLE-ABS-KEY ( "multi agent reinforcement learning" OR "multiagent reinforcement learning" OR marl OR "decentrali?ed reinforcement learning" OR "stochastic game*" OR "markov game*" OR "agent based simulat*" ) AND ALL ( "preference vector*" OR "preference weight*" OR "weight vector*" OR "preference condition*" OR "multi objective reinforcement learning" OR morl OR scalari?ation OR "reward weight*" ) AND ALL ( consensus OR "collective decision*" OR "group decision*" OR "emergent agreement" OR "joint policy" OR "joint policies" OR "policy aggregation" OR coordinat* OR "social welfare" ) AND ( LIMIT-TO ( SRCTYPE , "j" ) ) AND ( LIMIT-TO ( DOCTYPE , "ar" ) OR LIMIT-TO ( DOCTYPE , "re" ) )
```

The results of the application of the above statement in Scopus, filtered to include only Journals as Source type, and Article and Review as Document Type, are presented in Figure 1.

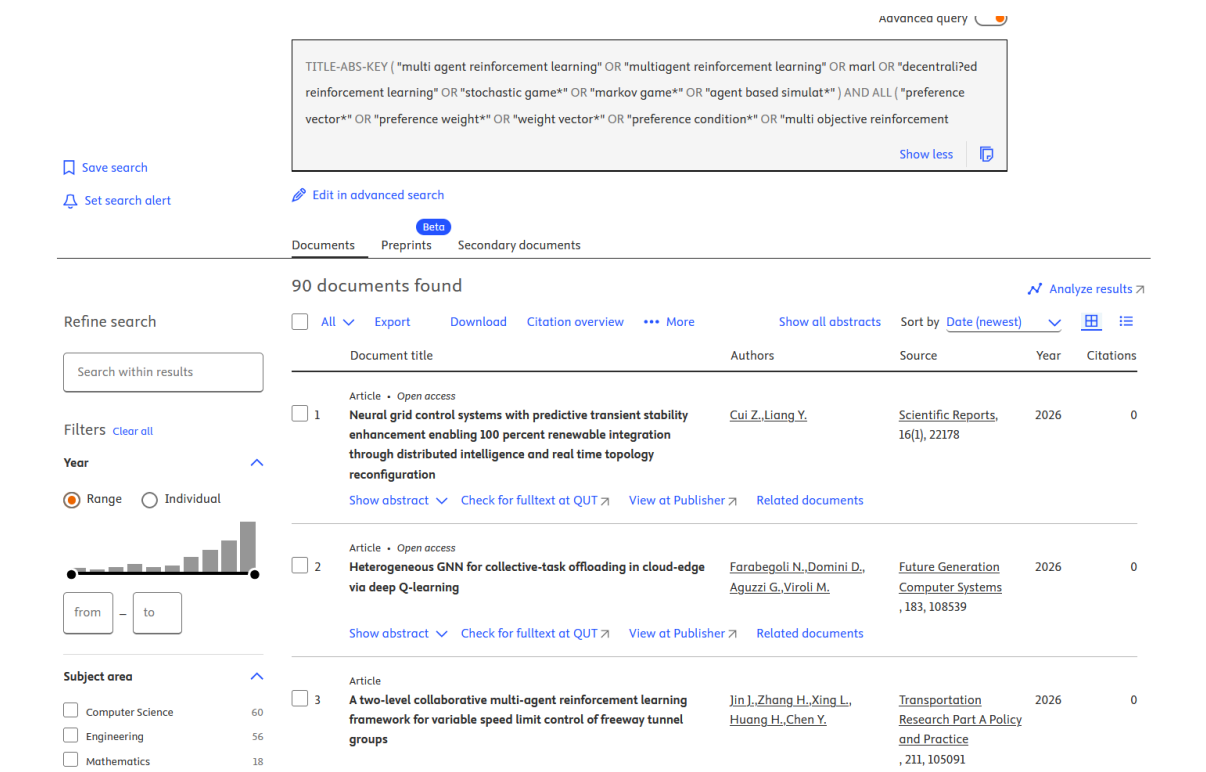


Figure 1. Search results

4.2. Search strategy development

The final search statement is built iteratively, starting from the most restrictive search and then expanding it to a reasonable number of results. The first search statement outputs only three articles (see Figure 2). Even though they are from the right disciplines, there are still too few to be considered a good search. Therefore, some of the restrictions of the statement (i.e. AND Boolean operators and restricted field search) need to change. From the three research concepts, the most restricted one is consensus. That is because, usually, computer science and mathematics are interrelated, and they show up in several resources together. Therefore, the most restrictive term is consensus, or the research concept where consensus or unanimity is mentioned. That is because being a decision-making system makes it less likely to show up together with computer science and mathematics. That is where all the work needs to focus to improve our search, at least in the first iterations.

Advanced query

(TITLE-ABS-KEY (("MARL simulation" OR MARL) OR (Multi-agent AND (reinforce* OR simulat*)) OR Agent* OR (Markov AND games OR chain*)) AND TITLE-ABS-KEY (Parametric AND (vector OR "preference vector*" OR weight*) OR ("preference-condition*" AND polic*) OR (MORL OR ("multi-objective reinforce*" AND learn*) OR scalarisation OR "reward

Show less

Save search Set search alert Edit in advanced search

Documents Preprints Secondary documents

3 documents found Analyze results

Refine search Search within results

Filters

Year

Subject area

Computer Science 2

Mathematics 2

Decision Sciences 1

Document type

Document title	Authors	Source	Year	Citations
1 A knowledge-informed multi-agent reinforcement learning approach for cost-effective aesthetic facades generative design	Wang B., Lu W., Xu Z., An H.	Advanced Engineering Informatics, 74, 104743	2026	0
2 Semiparametric model averaging prediction: a Bayesian approach	Wang J., Li J.	Australian and New Zealand Journal of Statistics, 60(4)	2018	3

Show abstract Check for fulltext at QUT View at Publisher Related documents

Figure 2. First iteration

Changing the last part to ALL fields instead of just TITLE-ABS-KEY produces two results (see Figure 3).

Advanced query

TITLE-ABS-KEY ("multi agent reinforcement learning" OR "multiagent reinforcement learning" OR marl OR "stochastic game*" OR "markov game*" OR "agent based simulat*" OR "agent based model*" OR "decentralized reinforcement learning") AND TITLE-ABS-KEY ("preference vector*" OR "preference weight*" OR "weight vector*" OR "preference

Show less

Save search Set search alert Edit in advanced search

Documents Preprints Secondary documents

2 documents found Analyze results

Refine search Search within results

Filters

Year

Range Individual

from to

Subject area

Computer Science 2

Mathematics 2

Engineering 1

Document title	Authors	Source	Year	Citations
1 ET-MATD3-based UAV formation control in dynamic obstacle environments	Li J., Liu L., Liang C., Yan D., Li L.	Journal of the Franklin Institute, 363(9), 108729	2026	0
2 Balancing Act: Prioritization Strategies for LLM-Designed Restless Bandit Rewards	Verma S., Boehmer N., Kong L., Tambe M.	Lecture Notes in Computer Science, 16223 LNCS	2026	2

Show abstract Check for fulltext at QUT View at Publisher Related documents

Display 10 results Back to top

Figure 3. Second iteration

When changing the third sub-search statement that handles the consensus and unanimity terms, always gives few results. Therefore, it is the Consensus research concept that is restricting the search. This hypothesis needs to be tested. If true, it means that the consensus search term requires better similar terms and synonyms.

When removing the third concept from the search statement, the Figure 4 shows the results increase to seventy-seven. This confirms the hypothesis. Therefore, the refined table is the following

Research concept	Keyword	Synonyms	Related terms
MARL simulation	Multi-agent reinforcement learning (MARL)	multi-agent RL; decentralised reinforcement learning	agent-based simulation; Markov/stochastic games; multi-agent systems
Parametric preference vectors	Preference vector	preference weights; preference-conditioned policy	multi-objective reinforcement learning (MORL); scalarisation; reward weighting
Consensus output prediction	Consensus	collective decision; group decision	emergent agreement joint policy; policy aggregation; coordination; social welfare

Figure 4. Refined table

Using the same strategy described in the previous section provides the final search, which returns ninety good-quality results. All those results belong to the disciplines that this report considers important. Therefore, this is deemed the final search statement (see section 4.1).

4.3. Search results analysis

The criteria that are used to evaluate the results are the following: First, it evaluates the quality of the results. Second, it assesses the scope of the results. And third, it evaluates whether the results belong to the disciplines that are relevant to the research question and the project itself.

The ninety results obtained from the search statement are already filtered to show only journals that are peer-reviewed, and they include only articles and reviews. Therefore, the quality of the results is ensured by showing only articles from peer-reviewed journals.

The scope of the results is evaluated based on the years of publication and the range of the whole set of articles that the search outputs. The years of publication range from 2017 to 2026, which reflects the fact that this multidisciplinary field is recent. Only when the graphics processing units, GPUs, were developed, they made multi-agent reinforcement learning, possible. That is the starting point of this specific field, and that occurred between 2015 and 2020, in the research labs. And this matches the dates of publication of the resulting articles.

Finally, the results' relevance is evaluated based on the number of disciplines that the research articles belong to. In this case, most of the articles belong to computer science, engineering, mathematics, social science and decision science, which are the core of the research. Even though most of the results belong to the fields this research question is focused on, there are seven articles that are from other unrelated disciplines. For example, medicine, earth and planetary sciences, biochemistry, genetics and molecular biology, and chemistry. These fields are filtered out.

5. Information management strategy

Due to the high number of research articles that the strategies output (i.e. ninety in this case), an information management strategy is fundamental for the success of a research project. For that reason, the following sections describe the steps taken to read, organize, store, classify, annotate, and retrieve the information in an efficient and organized way.

5.1. Reading and taking notes

All articles are filtered by reading the title, abstract and keywords first. This discards some of the irrelevant ones. Subsequently, the selected list is input into **Zotero**, including the actual articles' PDF.

The Zotero PDF reader is used locally. This reader allows writing and annotating the PDF and storing those comments and PDF text systematically for later referencing. Exporting references can be done in Zotero or in any other software tool.

5.2. Storing and organising literature

For storing and organizing literature, Zotero software is being used to store metadata (title, abstract, authors, DOI, etc.) and the PDF itself. The Zotero syncing feature is disabled to prevent the PDF being uploaded to unwanted servers.

Using Zotero's capabilities, the articles are classified according to their purpose: methodology, framework, application or review.

5.3. Managing references

Zotero is used as the primary reference manager using it plugging to Word text editor. This allows the direct referencing of in-text citations directly from Zotero. Since all articles are imported directly from Scopus, the whole process does not require manual data entry.

5.4. Documenting your searching

The search is documented using the Scopus built-in feature to save the search string and results (see the figure 5). This allows every search to be tracked with its results and date. Furthermore, in case extra comments need to be added, an .md file will be kept in OneDrive that links the search with the log-entry comments.



Saved searches

- 1 multi agent reinforcement lear [90 results](#) [🔗 New results](#) [More](#)
TITLE-ABS-KEY ("multi agent reinforcement learning" OR "multiagent reinforcement learning" OR marl OR "decentralized reinforcement learning" OR...
[Show more](#) [▼](#)

Figure 5. Saved searches

6. Current awareness strategy

In the Artificial Intelligence-related field, the advance of scientific knowledge has been staggering in the last decade. This makes an awareness strategy necessary to keep pace with cutting-edge AI science.

6.1. Strategy 1: Scopus alarms

The first awareness strategy is set using the Scopus built-in tool, where a periodic alarm sends the user an email at a set interval informing with the newest results for the user search statement. See figure 6.

Set search alert

×

TITLE-ABS-KEY ("multi agent reinforcement learning" ...
Show more ✓

Name of alert *

multi agent reinforcement learning multiagent reinfor

Email address *

oriolson.rodriquezramirez@hdr.qut.edu.au

Separate email addresses with a semicolon, comma, or space

Frequency

Every week ✓ on Friday ✓

Cancel

Set search alert

Figure 6. Scopus Alert

6.2. Strategy 2: Following lead researchers in LinkedIn.

The second strategy is to follow the top researchers in this interdisciplinary field. Usually, lead researchers publish on their LinkedIn pages when news has been released to the public. One of those researchers is Zhen Gao, who is the author of two of the articles resulting from the search. See figure 7.

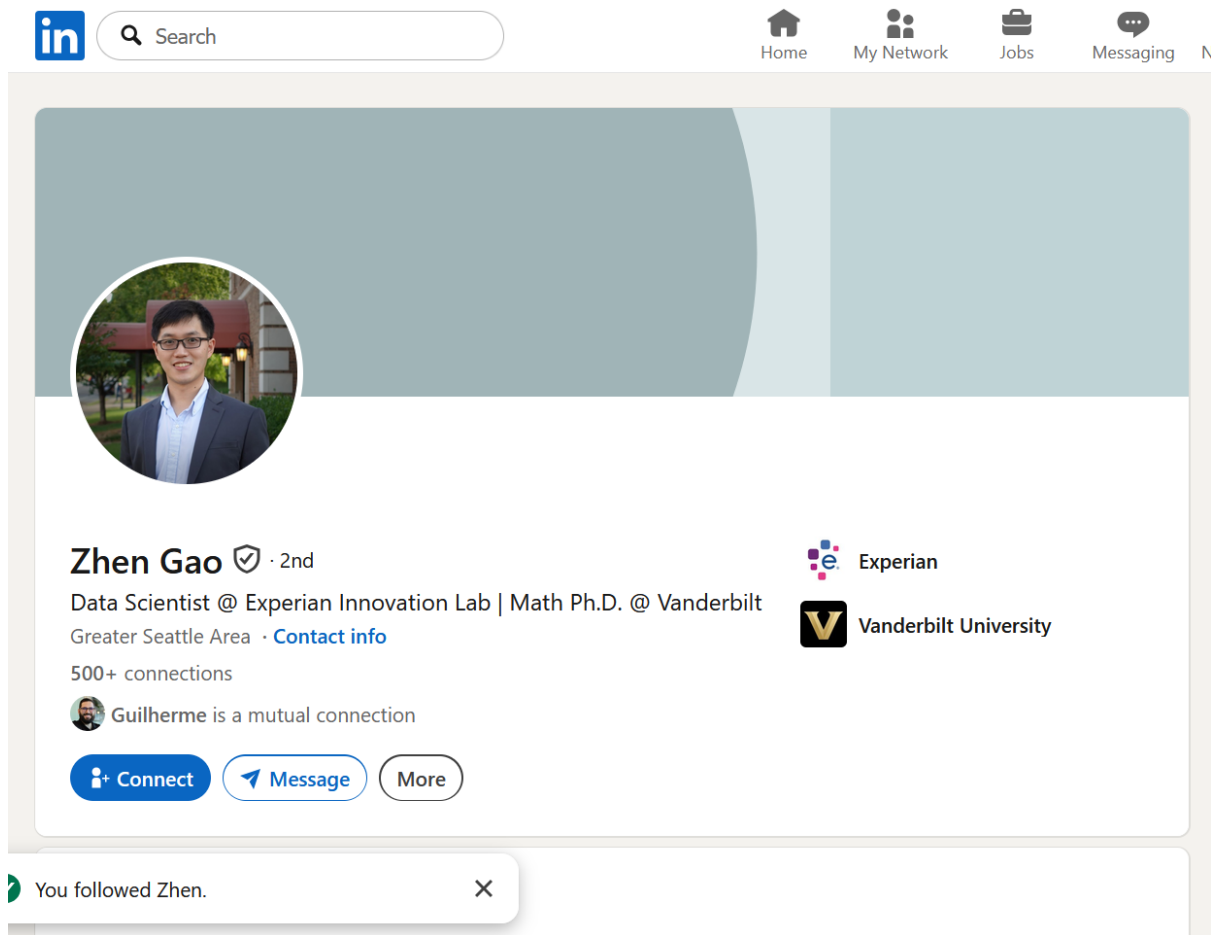


Figure 7. Top researcher

6.3. Strategy 3: Conference attendance and subscription

In this section, three provided references are analysed for authenticity following a structured validation strategy. For each source, a judgement of authenticity is provided, along with the process that leads to that judgement. When the source's judgement resolves to "fabricated", no other criterion is used, since there is no point in continuing.

7. Sample source analysis

In this section three provided references are analysed for authenticity following a structured validation strategy. For each source a judgement of authenticity is provided and the process that leads to that judgement. When the source's judgement resolves to "fabricated", no other criteria are used, since there is no point in continuing.

7.1. Source 1

Gunton, L. & Thorpe, C. (2023). Academic integrity as a shared institutional responsibility: The evolving role of librarians in higher education. *Journal of the Australian Library and Information Association*, 72(3), Article 12.

<https://doi.org/10.1080/24750158.2025.2460128>

Authenticity judgement:

Fabricated

Verification process and criteria:

- Does the metadata make sense? No, the DOI shows "2025" but the year of publication is 2023. This mismatch is a red flag.
- Does the DOI resolve to the referenced article? No, the given DOI leads to a different article and different authors.

7.2. Source 2

Walter, M. (2025). Doing Indigenous statistics in Australia: The racial burden of disregard. In Chris Andersen, Maggie Walter, Tahu Kukutai, & Chelsea Gabel (Eds.), *Indigenous statistics: From data deficits to data sovereignty*. (2nd ed., pp. 139–159). Routledge.

Authenticity judgement:

Genuine

Verification process and criteria:

- Is the resource found when searching the full citation string in Google scholar? Yes.
- Do the authors match the resource found on the internet? Yes.
- Do the referenced pages match the book's pages? Yes (the book is open access).
- Does the referenced chapter match the one in the book? Yes.

7.3. Source 3

Nourizadeh, P., Milford, M., & Fischer, T. (2024, May 13-17). Teach and repeat navigation: A robust control approach [Conference paper]. 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, pp. 2909-2916.

<https://doi.org/10.1109/ICRA57147.2024.10611662>

Authenticity judgement:

Genuine

Verification process and criteria:

- Does the metadata make sense? Yes, there is no contradictory information in the reference metadata.
- Does the DOI resolve to an article with the same metadata as the reference? Yes.
- Do the authors match the resource found on the internet? Yes.
- Do the referenced pages match the paper's pages? Yes.
- Is the resource found when doing a string Google scholar search? Yes.

- Does the journal or conference match the reference? Partly: the reference shows "[Conference paper]" and the actual conference is "2024 IEEE International Conference on Robotics and Automation (ICRA)", but that does not make it fabricated, just not complete enough.